

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define Blockchain. List out the advantage of Cryptocurrency.	Understand	6
B) Compare hash pointer and Markle tree and its application in blockchain security.	Understand	6
C) Explain Zero-knowledge system.	Understand	6
Q.2 Solve Any Two of the following.		12
A) What is double spending? How to avoid it in Bitcoin transaction?	Understand	6
B) Compare Bank system and Bitcoin System.	Analysis	6
C) Explain in detail proof of work (PoW).	Understand	6
Q. 3 Solve Any Two of the following.		12
A) Explain the Raft Algorithm in detail.	Understand	6
B) What are the design issues for the permission blockchains?	Understand	6
C) Explain Byzantine General Problem.	Understand	6
Q.4 Solve Any Two of the following.		12
A) Express the features of blockchain and the global economy of cross-border payments.	Apply	6
B) Elaborate on the idea behind KYC and its challenges, problems, and solutions.	Understand	6
C) Describe blockchain mortgage in detail.	Understand	6
Q. 5 Solve Any Two of the following.		12
A) Describe about Hyperledger fabric architecture.	Understand	6
B) Explain in detail about Corda.	Understand	6
C) Write a difference between ripple and ethereum?	Analysis	6